

TMS320x2833x Analog-to-Digital Converter (ADC) Module

Reference Guide

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Read This First

Notational Conventions

This document uses the following conventions.

- Hexadecimal numbers are shown with the suffix h or with a leading 0x. For example, the following number is 40 hexadecimal (decimal 64): 40h or 0x40.
- Registers in this document are shown in figures and described in tables.
 - Each register figure shows a rectangle divided into fields that represent the fields of the register. Each field is labeled with its bit name, its beginning and ending bit numbers above, and its read/write properties below. A legend explains the notation used for the properties.
 - Reserved bits in a register figure designate a bit that is used for future device expansion.

Related Documents From Texas Instruments

The following documents are available for download from the Texas Instruments website, www.ti.com.

Data Manual—

SPRS439— [TMS320F28335, F28334, F28332 Digital Signal Controllers \(DSCs\) Data Manual](#) contains the pinout, signal descriptions, as well as electrical and timing specifications for the F2833x devices.

CPU User's Guides—

SPRU430— [TMS320C28x DSP CPU and Instruction Set Reference Guide](#) describes the central processing unit (CPU) and the assembly language instructions of the TMS320C28x fixed-point digital signal processors (DSPs). It also describes emulation features available on these DSPs.

SPRUE02— [TMS320C28x Floating Point Unit and Instruction Set Reference Guide](#) describes the floating-point unit and includes the instructions for the FPU.

Peripheral Guides—

SPRU566— [TMS320x28xx, 28xxx Peripheral Reference Guide](#) describes the peripheral reference guides of the 28x digital signal processors (DSPs).

SPRUFB0— [TMS320x2833x System Control and Interrupts Reference Guide](#) describes the various interrupts and system control features of the 2833x digital signal controllers (DSCs).

SPRU812— [TMS320x2833x Analog-to-Digital Converter \(ADC\) Reference Guide](#) describes how to configure and use the on-chip ADC module, which is a 12-bit pipelined ADC.

SPRU949— [TMS320x2833x External Interface \(XINTF\) User's Guide](#) describes the XINTF, which is a nonmultiplexed asynchronous bus, as it is used on the 2833x devices.

SPRU963— [TMS320x2833x Boot ROM User's Guide](#) describes the purpose and features of the bootloader (factory-programmed boot-loading software) and provides examples of code. It also describes other contents of the device on-chip boot ROM and identifies where all of the information is located within that memory.

SPRUFB7— [TMS320x2833x Multichannel Buffered Serial Port \(McBSP\) User's Guide](#) describes the McBSP available on the F2833x devices. The McBSPs allow direct interface between a DSP and other devices in a system.

SPRUFB8— [TMS320x2833x Direct Memory Access \(DMA\) Reference Guide](#) describes the DMA on the 2833x devices.

- SPRU791—** [TMS320x28xx, 28xxx Enhanced Pulse Width Modulator \(ePWM\) Module Reference Guide](#) describes the main areas of the enhanced pulse width modulator that include digital motor control, switch mode power supply control, UPS (uninterruptible power supplies), and other forms of power conversion.
- SPRU924—** [TMS320x28xx, 28xxx High-Resolution Pulse Width Modulator \(HRPWM\)](#) describes the operation of the high-resolution extension to the pulse width modulator (HRPWM).
- SPRU807—** [TMS320x28xx, 28xxx Enhanced Capture \(eCAP\) Module Reference Guide](#) describes the enhanced capture module. It includes the module description and registers.
- SPRU790—** [TMS320x28xx, 28xxx Enhanced Quadrature Encoder Pulse \(eQEP\) Reference Guide](#) describes the eQEP module, which is used for interfacing with a linear or rotary incremental encoder to get position, direction, and speed information from a rotating machine in high performance motion and position control systems. It includes the module description and registers.
- SPRU074—** [TMS320x28xx, 28xxx Enhanced Controller Area Network \(eCAN\) Reference Guide](#) describes the eCAN that uses established protocol to communicate serially with other controllers in electrically noisy environments.
- SPRU051—** [TMS320x28xx, 28xxx Serial Communication Interface \(SCI\) Reference Guide](#) describes the SCI, which is a two-wire asynchronous serial port, commonly known as a UART. The SCI modules support digital communications between the CPU and other asynchronous peripherals that use the standard non-return-to-zero (NRZ) format.
- SPRU059—** [TMS320x28xx, 28xxx Serial Peripheral Interface \(SPI\) Reference Guide](#) describes the SPI - a high-speed synchronous serial input/output (I/O) port - that allows a serial bit stream of programmed length (one to sixteen bits) to be shifted into and out of the device at a programmed bit-transfer rate.
- SPRU721—** [TMS320x28xx, 28xxx Inter-Integrated Circuit \(I2C\) Reference Guide](#) describes the features and operation of the inter-integrated circuit (I2C) module that is available on the TMS320x280x digital signal processor (DSP).
- Tools Guides—**
- SPRU513—** [TMS320C28x Assembly Language Tools User's Guide](#) describes the assembly language tools (assembler and other tools used to develop assembly language code), assembler directives, macros, common object file format, and symbolic debugging directives for the TMS320C28x device.
- SPRU514—** [TMS320C28x Optimizing C Compiler User's Guide](#) describes the TMS320C28x™ C/C++ compiler. This compiler accepts ANSI standard C/C++ source code and produces TMS320 DSP assembly language source code for the TMS320C28x device.
- SPRU608—** [The TMS320C28x Instruction Set Simulator Technical Overview](#) describes the simulator, available within the Code Composer Studio for TMS320C2000 IDE, that simulates the instruction set of the C28x™ core.
- SPRU625—** [TMS320C28x DSP/BIOS Application Programming Interface \(API\) Reference Guide](#) describes development using DSP/BIOS.

Analog-to-Digital Converter (ADC)

The TMS320x2833x ADC module is a 12-bit pipelined analog-to-digital converter (ADC). The analog circuits of this converter, referred to as the core in this document, include the front-end analog multiplexers (MUXs), sample-and-hold (S/H) circuits, the conversion core, voltage regulators, and other analog supporting circuits. Digital circuits, referred to as the wrapper in this document, include programmable conversion sequencer, result registers, interface to analog circuits, interface to device peripheral bus, and interface to other on-chip modules.

This reference guide is applicable for the ADC found on the TMS320x2833x family of processors. This includes all Flash-based, ROM-based and RAM-based devices within the 2833x family.

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1.1 Features

The ADC module has 16 channels, configurable as two independent 8-channel modules to service the ePWM modules. The two independent 8-channel modules can be cascaded to form a 16-channel module. Although there are multiple input channels and two sequencers, there is only one converter in the ADC module. [Figure 1-1](#) shows the block diagram of the ADC module.

The two 8-channel modules can autosequence a series of conversions; each module has the choice of selecting any one of the respective eight channels available through an analog MUX. In the cascaded mode, the autosequencer functions as a single 16-channel sequencer. On each sequencer, once the conversion is completed, the selected channel value is stored in its respective ADCRESULT register. Autosequencing allows the system to convert the same channel multiple times, allowing the user to perform oversampling algorithms. This oversampling gives increased resolution over traditional single-sampled conversion results.

Functions of the ADC module include:

- 12-bit ADC core with built-in dual sample-and-hold (S/H)
- Simultaneous sampling or sequential sampling modes
- Analog input: 0 V to 3 V
- Fast conversion time runs at 12.5 MHz, ADC clock, or 6.25 MSPS
- 16-channel, multiplexed inputs
- Autosequencing capability provides up to 16 "autoconversions" in a single session. Each conversion can be programmed to select any 1 of 16 input channels.
- Sequencer can be operated as two independent 8-state sequencers or as one large 16-state sequencer (i.e., two cascaded 8-state sequencers).
- Sixteen result registers (individually addressable) to store conversion values
 - The digital value of the input analog voltage is derived by:

$$\text{Digital Value} = 0, \quad \text{when input} \leq 0 \text{ V}$$

$$\text{Digital Value} = 4096 \times \frac{\text{Input Analog Voltage} - \text{ADCLO}}{3} \quad \text{when } 0 \text{ V} < \text{input} < 3 \text{ V}$$

$$\text{Digital Value} = 4095, \quad \text{when input} \geq 3 \text{ V}$$

A All fractional values are truncated.

- Multiple triggers as sources for the start-of-conversion (SOC) sequence
 - S/W - software immediate start
 - ePWM 1-6
 - GPIO XINT2
- Flexible interrupt control allows interrupt request on every end-of-sequence (EOS) or every other EOS
- Sequencer can operate in "start/stop" mode, allowing multiple "time-sequenced triggers" to synchronize conversions.
- ePWM triggers can operate independently in dual-sequencer mode.
- Sample-and-hold (S/H) acquisition time window has separate prescale control.

Figure 1-1. Block Diagram of the ADC Module

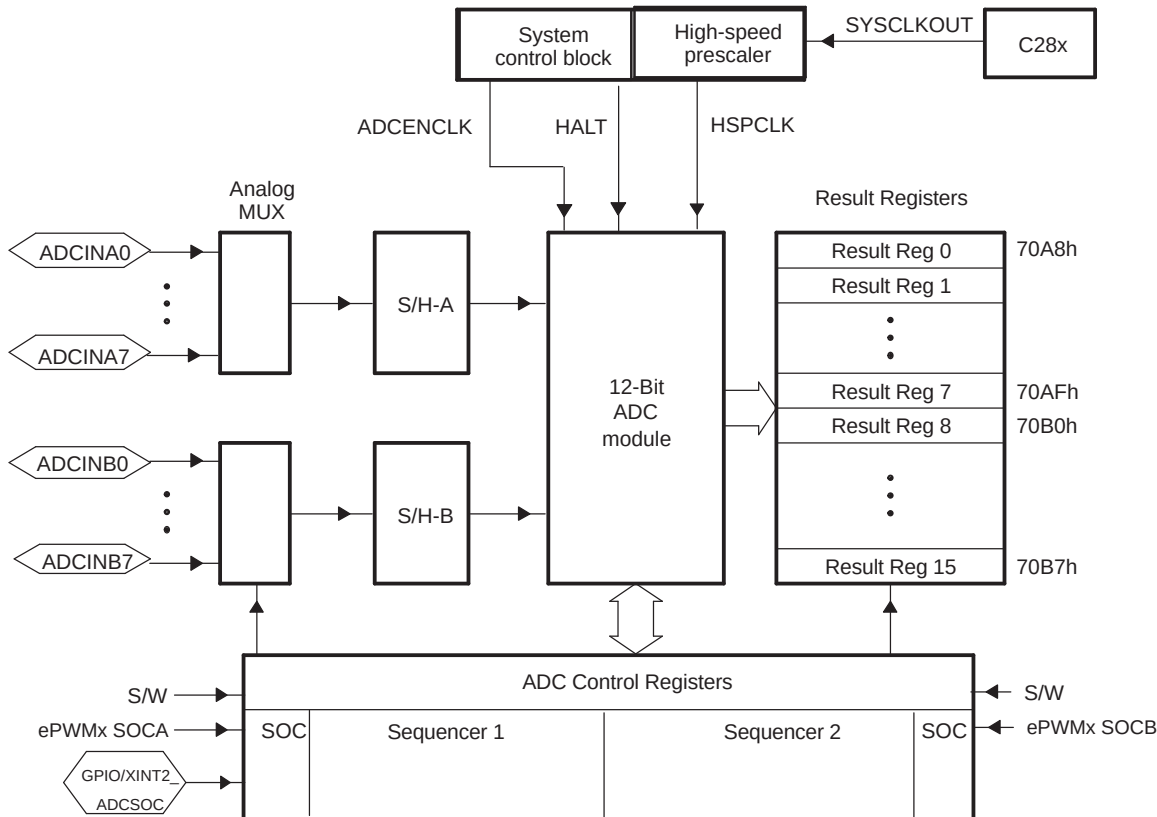


Table 1-1. ADC Registers

Name	Address ⁽¹⁾	Address ⁽²⁾	Size (x16)	Description
ADCTRL1	0x7100		1	ADC Control Register 1
ADCTRL2	0x7101		1	ADC Control Register 2
ADCMAxCONV	0x7102		1	ADC Maximum Conversion Channels Register
ADCCHSELSEQ1	0x7103		1	ADC Channel Select Sequencing Control Register 1
ADCCHSELSEQ2	0x7104		1	ADC Channel Select Sequencing Control Register 2
ADCCHSELSEQ3	0x7105		1	ADC Channel Select Sequencing Control Register 3
ADCCHSELSEQ4	0x7106		1	ADC Channel Select Sequencing Control Register 4
ADCASEQSR	0x7107		1	ADC Auto-Sequence Status Register
ADCRESULT0	0x7108	0x0B00	1	ADC Conversion Result Buffer Register 0
ADCRESULT1	0x7109	0x0B01	1	ADC Conversion Result Buffer Register 1
ADCRESULT2	0x710A	0x0B02	1	ADC Conversion Result Buffer Register 2
ADCRESULT3	0x710B	0x0B03	1	ADC Conversion Result Buffer Register 3
ADCRESULT4	0x710C	0x0B04	1	ADC Conversion Result Buffer Register 4
ADCRESULT5	0x710D	0x0B05	1	ADC Conversion Result Buffer Register 5
ADCRESULT6	0x710E	0x0B06	1	ADC Conversion Result Buffer Register 6
ADCRESULT7	0x710F	0x0B07	1	ADC Conversion Result Buffer Register 7
ADCRESULT8	0x7110	0x0B08	1	ADC Conversion Result Buffer Register 8
ADCRESULT9	0x7111	0x0B09	1	ADC Conversion Result Buffer Register 9

⁽¹⁾ The registers in this column are Peripheral Frame 2 registers.

⁽²⁾ The ADC result registers are dual mapped in the device. Locations in Peripheral Frame 2 (0x7108-0x7117) are 2 wait states and left justified. Locations in Peripheral Frame 0 space (0x0B00-0x0B0F) are 0 wait states and right justified. During high speed/continuous conversion use of the ADC, use the 0 wait state locations to avoid missing ADC conversions.

Table 1-1. ADC Registers (continued)

Name	Address ⁽¹⁾	Address ⁽²⁾	Size (x16)	Description
ADCRESLT10	0x7112	0x0B0A	1	ADC Conversion Result Buffer Register 10
ADCRESLT11	0x7113	0x0B0B	1	ADC Conversion Result Buffer Register 11
ADCRESLT12	0x7114	0x0B0C	1	ADC Conversion Result Buffer Register 12
ADCRESLT13	0x7115	0x0B0D	1	ADC Conversion Result Buffer Register 13
ADCRESLT14	0x7116	0x0B0E	1	ADC Conversion Result Buffer Register 14
ADCRESLT15	0x7117	0x0B0F	1	ADC Conversion Result Buffer Register 15
ADCTRL3	0x7118		1	ADC Control Register 3
ADCST	0x7119		1	ADC Status Register
Reserved	0x711A 0x711B		2	
ADCREFSEL	0x711C		1	ADC Reference Select Register
ADCOFFTRIM	0x711D		1	ADC Offset Trim Register
Reserved	0x711E 0x711F		2	ADC Status Register

To obtain the specified accuracy of the ADC, proper board layout is very critical. To the best extent possible, traces leading to the ADCINxx pins should not run in close proximity to the digital signal paths. This is to minimize switching noise on the digital lines from getting coupled to the ADC inputs. Furthermore, proper isolation techniques must be used to isolate the ADC module power pins from the digital supply.

1.2 Autoconversion Sequencer Principle of Operation

The ADC sequencer consists of two independent 8-state sequencers (SEQ1 and SEQ2) that can also be cascaded together to form one 16-state sequencer (SEQ). The word "state" represents the number of autoconversions that can be performed with the sequencer. Block diagrams of the single (16-state, cascaded) and dual (two 8-state, separated) sequencer modes are shown in [Figure 1-4](#) and [Figure 1-5](#), respectively.

In both cases, the ADC has the ability to autosequence a series of conversions. This means that each time the ADC receives a start-of-conversion request, it can perform multiple conversions automatically. For every conversion, any one of the available 16 input channels can be selected through the analog MUX. After conversion, the digital value of the selected channel is stored in the appropriate result register (ADCRESLTn). (The first result is stored in ADCRESULT0, the second result in ADCRESULT1, and so on). It is also possible to sample the same channel multiple times, allowing the user to perform "over-sampling", which gives increased resolution over traditional single-sampled conversion results.

Note: In the sequential sampling dual-sequencer mode, a pending SOC request from either sequencer is taken up as soon as the sequence initiated by the currently active sequencer is finished. For example, assume that the A/D converter is busy catering to SEQ2 when an SOC request from SEQ1 occurs. The A/D converter will start SEQ1 immediately after completing the request in progress on SEQ2. If SOC requests are pending from both SEQ1 and SEQ2, the SOC for SEQ1 has priority. For example, assume that the A/D converter is busy catering to SEQ1. During that process, SOC requests from both SEQ1 and SEQ2 are made. When SEQ1 completes its active sequence, the SOC request for SEQ1 will be taken up immediately. The SOC request for SEQ2 will remain pending.

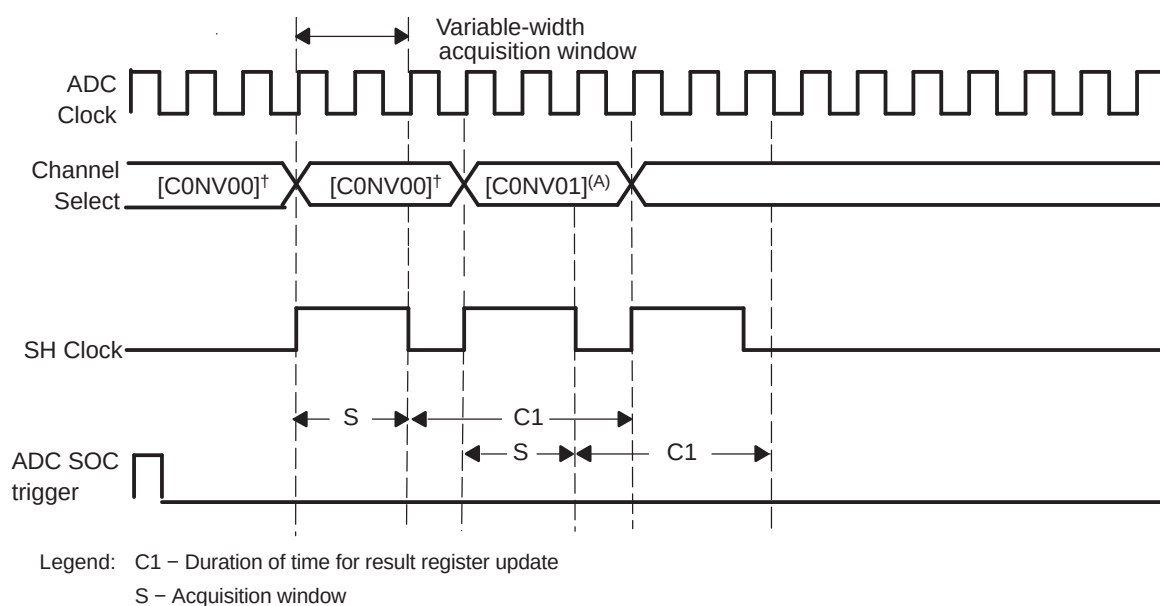
The ADC can also operate in simultaneous sampling mode or sequential sampling mode. For each conversion (or pair of conversions in simultaneous sampling mode), the current CONVxx bit field defines the pin (or pair of pins) to be sampled and converted. In sequential sampling mode, all four bits of CONVxx define the input pin. The MSB defines which sample-and-hold buffer the input pin is associated with, and the three LSBs define the offset. For example, if CONVxx contains the value 0101b, ADCINA5 is the selected input pin. If it contains the value 1011b, ADCINB3 is the selected input pin. In simultaneous

sampling mode, the MSB of the CONVxx register is discarded. Each sample and hold buffer samples the associated pin given by the offset provided in the three LSBs of the CONVxx register. For instance, if the CONVxx register contains the value 0110b, ADCINA6 is sampled by S/H-A and ADCINB6 is sampled by S/H-B. If the value is 1001b, ADCINA1 is sampled by S/H-A and ADCINB1 is sampled by S/H-B. The voltage in S/H-A is converted first, followed by the S/H-B voltage. The result of the S/H-A conversion is placed in the current ADCRESULTn register (ADCRESULT0 for SEQ1, assuming the sequencer has been reset). The result of the S/H-B conversion is placed in the next ADCRESULTn register (ADCRESULT1 for SEQ1, assuming the sequencer has been reset). The result register pointer is then increased by two (to point to ADCRESULT2 for SEQ1, assuming the sequencer had originally been reset).

1.2.1 Sequential Sampling Mode

Figure 1-2 shows the timing of sequential sampling mode. In this example, the ACQ_PS bits are set to 0001b.

Figure 1-2. Sequential Sampling Mode (SMODE = 0)

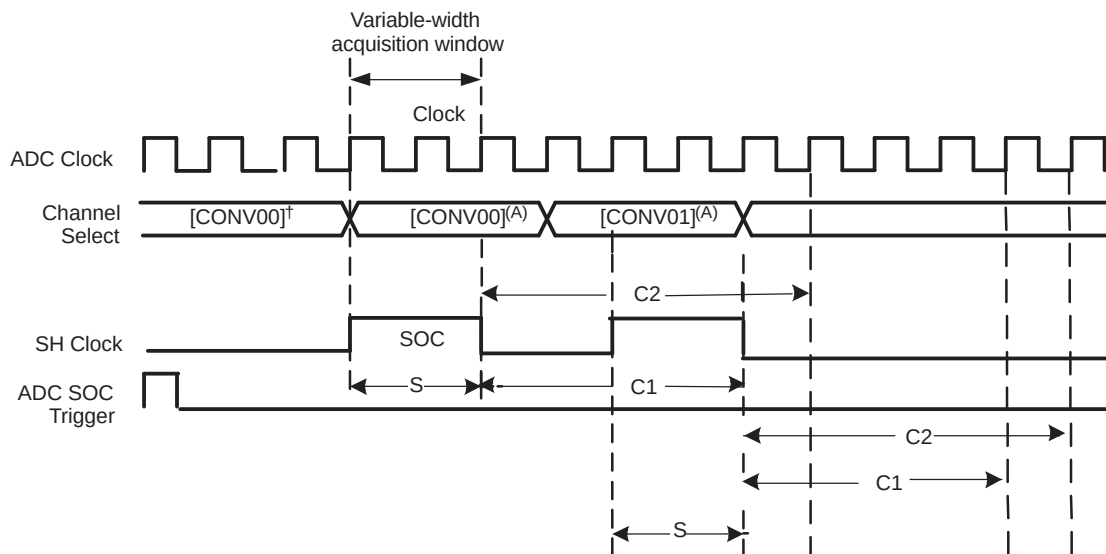


A ADC channel address contained in [CONV00] 4-bit register; CONV00 for SEQ1 and CONV08 for SEQ2.

1.2.2 Simultaneous Sampling Mode

Figure 1-3 describes the timing of simultaneous sampling mode. In this example, the ACQ_PS bits are set to 0001b.

Figure 1-3. Simultaneous Sampling Mode (SMODE=1)



Legend: C1 – Duration of time for Ax channel result in result register

C2 – Duration of time for Bx channel result in result register

S – Acquisition window

- A ADC channel address contained in [CONV00] 4-bit register; [CONV00] means A0/B0 channels; [CONV01] means A1/B1 channels.

Figure 1-4. Block Diagram of Autosequenced ADC in Cascaded Mode

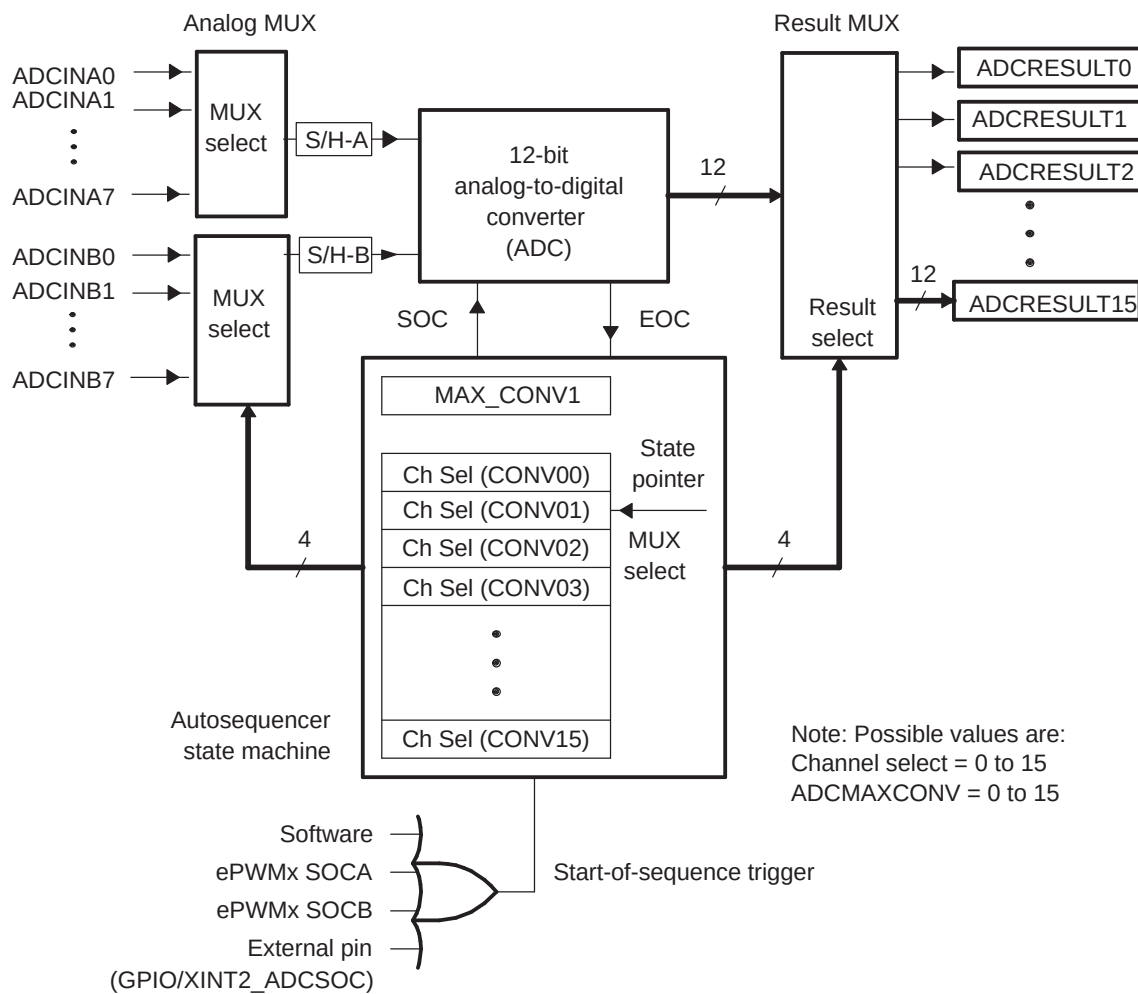
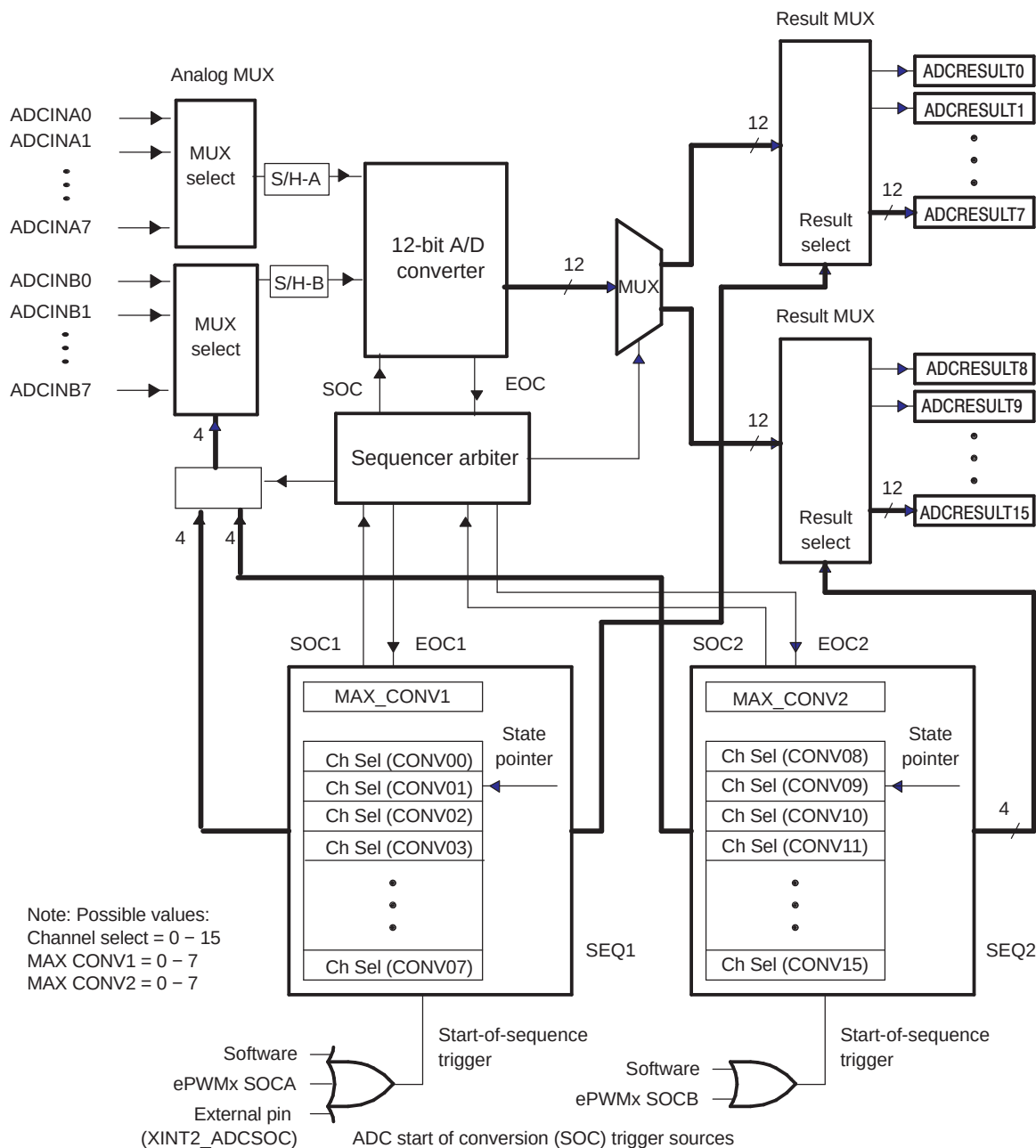


Figure 1-5. Block Diagram of Autosequenced ADC With Dual Sequencers



Note: There is only one ADC in the DSP. This converter is shared by the two sequencers in dual-sequencer mode.

The sequencer operation for both 8-state and 16-state modes is almost identical; the few differences are highlighted in [Table 1-2](#).

Table 1-2. Comparison of Single and Cascaded Operating Modes

Feature	Single 8-state sequencer #1 (SEQ1)	Single 8-state sequencer #2 (SEQ2)	Cascaded 16-state sequencer (SEQ)
Start-of-conversion (SOC) triggers	ePWMx SOCA, software, external pin	ePWMx SOCB, software	ePWMx SOCA, ePWMx SOCB, software, external pin
Maximum number of autoconversions (i.e., sequence length)	8	8	16
Autostop at end-of-sequence (EOS)	Yes	Yes	Yes
Arbitration priority	High	Low	Not applicable
ADC conversion result register locations	0 to 7	8 to 15	0 to 15
ADCCHSELSEQn bit field assignment	CONV00 to CONV07	CONV08 to CONV15	CONV00 to CONV15

For convenience, the sequencer states will be subsequently referred to as:

- For SEQ1: CONV00 to CONV07
- For SEQ2: CONV08 to CONV15
- For Cascaded SEQ: CONV00 to CONV15

The analog input channel selected for each sequenced conversion is defined by CONVxx bit fields in the ADC input channel select sequencing control registers (ADCCHSELSEQn). CONVxx is a 4-bit field that specifies any one of the 16 channels for conversion. Since a maximum of 16 conversions in a sequence is possible when using the sequencers in cascaded mode, 16 such 4-bit fields (CONV00 - CONV15) are available and are spread across four 16-bit registers (ADCCHSELSEQ1 - ADCCHSELSEQ4). The CONVxx bits can have any value from 0 to 15. The analog channels can be chosen in any desired order and the same channel may be selected multiple times.

Example 1-1. Simultaneous Sampling Dual Sequencer Mode Example

Example initialization:

```

AdcRegs.ADCCTRL3.bit.SMODE_SEL = 0x1;    // Setup simultaneous sampling mode
AdcRegs.ADCMAXCONV.all = 0x0033;         // 4 double conv's each sequencer (8 total)
AdcRegs.ADCCHSELSEQ1.bit.CONV00 = 0x0;   // Setup conv from ADCINA0 & ADCINB0
AdcRegs.ADCCHSELSEQ1.bit.CONV01 = 0x1;   // Setup conv from ADCINA1 & ADCINB1
AdcRegs.ADCCHSELSEQ1.bit.CONV02 = 0x2;   // Setup conv from ADCINA2 & ADCINB2
AdcRegs.ADCCHSELSEQ1.bit.CONV03 = 0x3;   // Setup conv from ADCINA3 & ADCINB3
AdcRegs.ADCCHSELSEQ3.bit.CONV08 = 0x4;   // Setup conv from ADCINA4 & ADCINB4
AdcRegs.ADCCHSELSEQ3.bit.CONV09 = 0x5;   // Setup conv from ADCINA5 & ADCINB5
AdcRegs.ADCCHSELSEQ3.bit.CONV10 = 0x6;   // Setup conv from ADCINA6 & ADCINB6
AdcRegs.ADCCHSELSEQ3.bit.CONV11 = 0x7;   // Setup conv from ADCINA7 & ADCINB7

```

If SEQ1 and SEQ2 were both executed, the results would go to the following RESULT registers:

```

ADCINA0 -> ADCRESULT0
ADCINB0 -> ADCRESULT1
ADCINA1 -> ADCRESULT2
ADCINB1 -> ADCRESULT3
ADCINA2 -> ADCRESULT4
ADCINB2 -> ADCRESULT5
ADCINA3 -> ADCRESULT6
ADCINB3 -> ADCRESULT7
ADCINA4 -> ADCRESULT8
ADCINB4 -> ADCRESULT9
ADCINA5 -> ADCRESULT10
ADCINB5 -> ADCRESULT11
ADCINA6 -> ADCRESULT12
ADCINB6 -> ADCRESULT13
ADCINA7 -> ADCRESULT14
ADCINB7 -> ADCRESULT15

```

Example 1-2. Simultaneous Sampling Cascaded Sequencer Mode Example

```

AdcRegs.ADCCTRL3.bit.SMODE_SEL    = 0x1;    // Setup simultaneous sampling mode
AdcRegs.ADCCTRL1.bit.SEQ_CASC      = 0x1;    // Setup cascaded sequencer mode
AdcRegs.ADCMAXCONV.all =           0x0007;    // 8 double conv's (16 total)
AdcRegs.ADCCHSELSEQ1.bit.CONV00    = 0x0;    // Setup conv from ADCINA0 & ADCINB0
AdcRegs.ADCCHSELSEQ1.bit.CONV01    = 0x1;    // Setup conv from ADCINA1 & ADCINB1
AdcRegs.ADCCHSELSEQ1.bit.CONV02    = 0x2;    // Setup conv from ADCINA2 & ADCINB2
AdcRegs.ADCCHSELSEQ1.bit.CONV03    = 0x3;    // Setup conv from ADCINA3 & ADCINB3
AdcRegs.ADCCHSELSEQ2.bit.CONV04    = 0x4;    // Setup conv from ADCINA4 & ADCINB4
AdcRegs.ADCCHSELSEQ2.bit.CONV05    = 0x5;    // Setup conv from ADCINA5 & ADCINB5
AdcRegs.ADCCHSELSEQ2.bit.CONV06    = 0x6;    // Setup conv from ADCINA6 & ADCINB6
AdcRegs.ADCCHSELSEQ2.bit.CONV07    = 0x7;    // Setup conv from ADCINA7 & ADCINB7

```

If the cascaded SEQ was executed, the results would go to the following ADCRESULT registers:

```

ADCINA0 -> ADCRESULT0
ADCINB0 -> ADCRESULT1
ADCINA1 -> ADCRESULT2
ADCINB1 -> ADCRESULT3
ADCINA2 -> ADCRESULT4
ADCINB2 -> ADCRESULT5
ADCINA3 -> ADCRESULT6
ADCINB3 -> ADCRESULT7
ADCINA4 -> ADCRESULT8
ADCINB4 -> ADCRESULT9
ADCINA5 -> ADCRESULT10
ADCINB5 -> ADCRESULT11
ADCINA6 -> ADCRESULT12
ADCINB6 -> ADCRESULT13
ADCINA7 -> ADCRESULT14
ADCINB7 -> ADCRESULT15

```

1.3 Uninterrupted Autosequenced Mode

The following description applies to the 8-state sequencers (SEQ1 or SEQ2). In this mode, SEQ1/SEQ2 can autosequence up to eight conversions of any channel in a single sequencing session (16 when sequencers are cascaded together). [Figure 1-6](#) shows the flow diagram. The result of each conversion is stored in one of the eight result registers (ADCRESULT0 - ADCRESULT7 for SEQ1 and ADCRESULT8 - ADCRESULT15 for SEQ2). These registers are filled from the lowest address to the highest address.

The number of conversions in a sequence is controlled by MAX_CONVn (a 3-bit or 4-bit field in the ADCMAXCONV register), which is automatically loaded into the sequencing counter status bits (SEQ_CNTR[3:0]) in the autosequence status register (ADCSEQSR) at the start of an autosequenced conversion session. The MAX_CONVn field can have a value ranging from zero to seven (zero to fifteen when sequencers are cascaded together). SEQ_CNTR bits count down from their loaded value as the sequencer starts from state CONV00 and continues sequentially (CONV01, CONV02, and so on) until SEQ_CNTR has reached zero. The number of conversions completed during an autosequencing session is equal to (MAX_CONVn + 1).

Example 1-3. Conversion in Dual-Sequencer Mode Using SEQ1

Suppose seven conversions are desired from SEQ1 (i.e., inputs ADCINA2 and ADCINA3 twice, then ADCINA6, ADCINA7, and ADCINA4 must be converted as part of the autosequenced session), then MAX_CONV1 should be set to 6 and the ADCCHSELSEQn registers should be set to the values shown in Table 1-3.

Conversion begins once the start-of-conversion (SOC) trigger is received by the sequencer. The SOC trigger also loads the SEQ_CNTR bits. Those channels that are specified in the ADCCHSELSEQn registers are taken up for conversion, in the predetermined sequence. The SEQ_CNTR bits are decremented by one automatically after every conversion. Once SEQ_CNTR reaches zero, two things can happen, depending on the status of the continuous run bit (CONT_RUN) in the ADCTRL1 register. See Figure 1-6 for an illustration of the flow.

- If CONT_RUN is set, the conversion sequence starts all over again automatically (i.e., SEQ_CNTR gets reloaded with the original value in MAX_CONV1 and SEQ1 state is set to CONV00 [See Section 1.7 for more options]). In this case, to avoid overwriting the data, you must be sure that the result registers are read before the next conversion sequence begins. The arbitration logic designed into the ADC ensures that the result registers are not corrupted should a contention arise (ADC module trying to write into the result registers while you try to read from them at the same time).
- If CONT_RUN is not set, the sequencer stays in the last state (CONV06, in this example) and SEQ_CNTR continues to hold a value of zero. To repeat the sequence on the next SOC, the sequencer must be reset using the RST_SEQn bit prior to the next SOC.

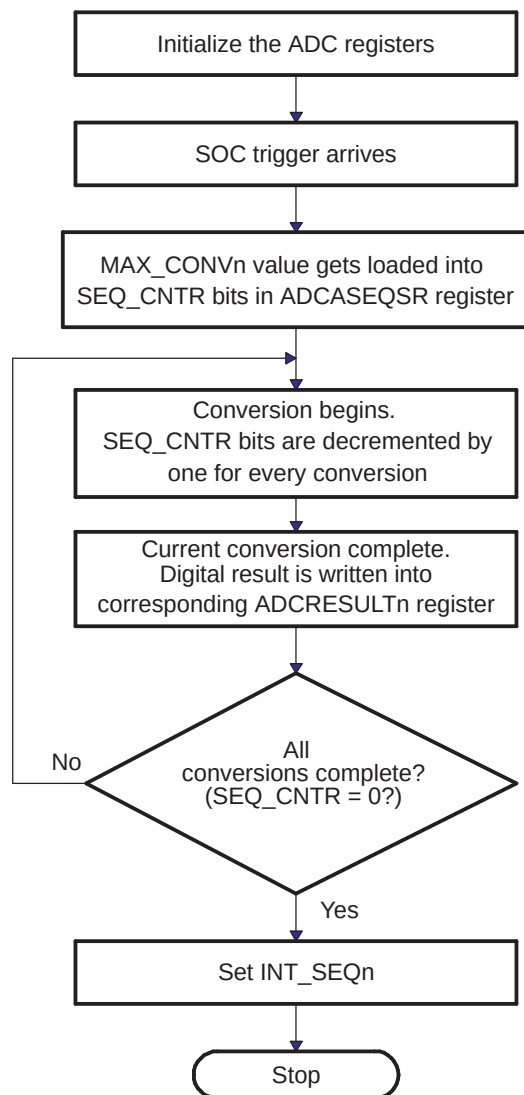
If the interrupt flag is set every time SEQ_CNTR reaches zero (INT_ENA_SEQn = 1 and INT_MOD_SEQn = 0), you can (if needed) manually reset the sequencer (using the RST_SEQn bit in the ADCTRL2 register) in the interrupt service routine (ISR). This causes the SEQn state to be reset to its original value (CONV00 for SEQ1 and CONV08 for SEQ2). This feature is useful in the Start/Stop operation of the sequencer. Example 1-3 also applies to SEQ2 and the cascaded 16-state sequencer (SEQ) with differences outlined in Table 1-2.

Table 1-3. Values for ADCCHSELSEQn Registers (MAX_CONV1 Set to 6)

	Bits 15-12 ⁽¹⁾	Bits 11-8 ⁽¹⁾	Bits 7-4 ⁽¹⁾	Bits 3-0 ⁽¹⁾	
70A3h	3	2	3	2	ADCCHSELSEQ1
70A4h	x	12	7	6	ADCCHSELSEQ2
70A5h	x	x	x	x	ADCCHSELSEQ3
70A6h	x	x	x	x	ADCCHSELSEQ4

⁽¹⁾ Values are in decimal, and x = don't care

Figure 1-6. Flow Chart for Uninterrupted Autosequenced Mode



A The flow chart corresponds to CONT_RUN bit = 0 and INT_MOD_SEQn bit = 0.

1.3.1 Sequencer Start/Stop Mode (Sequencer Start/Stop Operation With Multiple Time-Sequenced Triggers)

In addition to the uninterrupted autosequenced mode, any sequencer (SEQ1, SEQ2, or SEQ) can be operated in a Stop/Start mode which is synchronized to multiple start-of-conversion (SOC) triggers, separated in time. This mode is similar to [Example 1-3](#), but the sequencer is allowed to be retriggered without being reset to the initial state CONV00, once it has completed its first sequence (i.e., the sequencer is not reset in the interrupt service routine). Therefore, when one conversion sequence ends, the sequencer stays in the current conversion state. The continuous run bit (CONT_RUN) in the ADCTRL1 register must be set to zero (i.e., disabled) for this mode.

Example 1-4. Sequencer Start/Stop Operation

Requirement: To start three autoconversions (e.g., I_1, I_2, I_3) off trigger 1 (underflow) and three autoconversions (e.g., V_1, V_2, V_3) off trigger 2 (period). Triggers 1 and 2 are separated in time by 25 μs and are provided by an ePWM. See Figure 1-7. Only SEQ1 is used in this case.

Note: Triggers 1 and 2 may be an SOC signal from ePWM, external pin, or software. The same trigger source may occur twice to satisfy the dual-trigger requirement of this example. Care must be taken such that multiple ePWM triggers are not lost due sequences already in progress. See Section 1.3.3.

Here MAX_CONV1 is set to 2 and the ADC Input Channel Select Sequencing Control Registers (ADCCHSELSEQn) are set as shown in Table 1-4.

Once reset and initialized, SEQ1 waits for a trigger. With the first trigger, three conversions with channel-select values of: CONV00 (I_1), CONV01 (I_2), and CONV02 (I_3) are performed. SEQ1 then waits at current state for another trigger. Twenty-five microseconds later when the second trigger arrives, another three conversions occur, with channel-select values of CONV03 (V_1), CONV04 (V_2), and CONV05 (V_3).

The value of MAX_CONV1 is automatically loaded into SEQ_CNTR for both trigger cases. If a different number of conversions are required at the second trigger point, you must (at some appropriate time before the second trigger) change the value of MAX_CONV1 through software, otherwise, the current (originally loaded) value will be reused. This can be done by an ISR that changes the value of MAX_CONV1 at the appropriate time. The interrupt operation modes are described in Section 1.3.4.

At the end of the second autoconversion session, the ADC result registers will have the values shown in Table 1-5.

At this point, SEQ1 keeps "waiting" at the current state for another trigger. Now, the user can reset SEQ1 (by software) to state CONV00 and repeat the same trigger1, 2 sessions.

Figure 1-7. Example of ePWM Triggers to Start the Sequencer

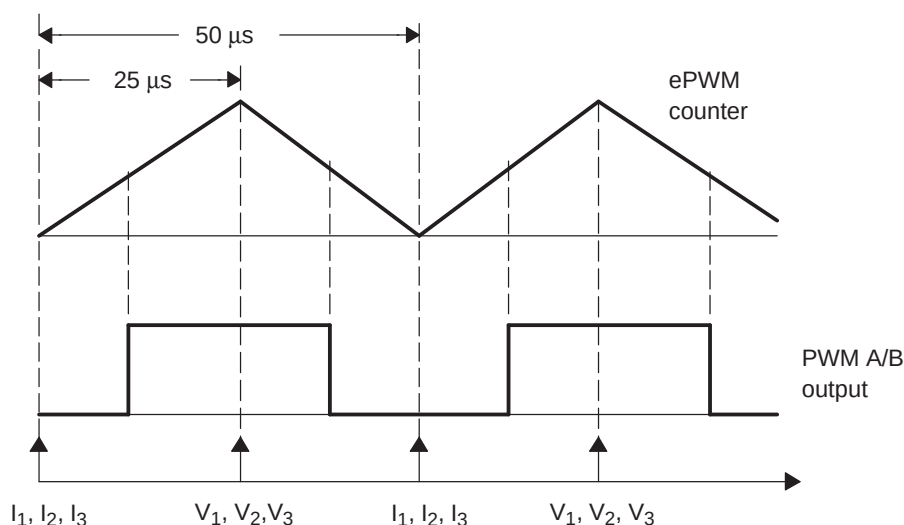


Table 1-4. Values for ADCCHSELSEQn (MAX_CONV1 set to 2)

	Bits 15-12	Bits 11-8	Bits 7-4	Bits 3-0	
70A3h	V ₁	I ₃	I ₂	I ₁	ADCCHSELSEQ1
70A4h	x	x	V ₃	V ₂	ADCCHSELSEQ2
70A5h	x	x	x	x	ADCCHSELSEQ3
70A6h	x	x	x	x	ADCCHSELSEQ4

Table 1-5. Values After Second Autoconversion Session

Buffer Register	ADC Conversion Result Buffer
ADCRESULT0	I ₁
ADCRESULT1	I ₂
ADCRESULT2	I ₃
ADCRESULT3	V ₁
ADCRESULT4	V ₂
ADCRESULT5	V ₃
ADCRESULT6	x
ADCRESULT7	x
ADCRESULT8	x
ADCRESULT9	x
ADCRESULT10	x
ADCRESULT11	x
ADCRESULT12	x
ADCRESULT13	x
ADCRESULT14	x
ADCRESULT15	x

1.3.2 Simultaneous Sampling Mode

The ADC has the ability to sample two ADCINxx inputs simultaneously, provided that one input is from the range ADCINA0 - ADCINA7 and the other input is from the range ADCINB0 - ADCINB7. Furthermore, the two inputs must have the same sample-and-hold offset (i.e., ADCINA4 and ADCINB4, but not ADCINA7 and ADCINB6). To put the ADC into simultaneous sampling mode, the SMODE_SEL bit in the ADCTRL3 register must be set. See [Section 1.2](#) for details.

1.3.3 Input Trigger Description

Each sequencer has a set of trigger inputs that can be enabled/disabled. See [Table 1-6](#) for the valid input triggers for SEQ1, SEQ2, and cascaded SEQ.

Table 1-6. Input Triggers

SEQ1 (sequencer 1)	SEQ2 (sequencer 2)	Cascaded SEQ
Software trigger (software SOC)	Software trigger (software SOC)	Software trigger (software SOC)
ePWMx SOCA	ePWMx SOCB	ePWMx SOCA
XINT2_ADCSOC		ePWMx SOCB
		XINT2_ADCSOC

Notes:

- An SOC trigger can initiate an autoconversion sequence whenever a sequencer is in an idle state. An idle state is either CONV00 prior to receiving a trigger, or any state which the sequencer lands on at the completion of a conversion sequence, i.e., when SEQ_CNTR has reached a count of zero.
 - If an SOC trigger occurs while a current conversion sequence is underway, it sets the SOC_SEQn bit (which would have been cleared on the commencement of a previous conversion sequence) in the ADCTRL2 register. If yet another SOC trigger occurs, it is lost (i.e., when the SOC_SEQn bit is already set (SOC pending), subsequent triggers will be ignored).
 - Once triggered, the sequencer cannot be stopped/halted in mid sequence. The program must either wait until an end-of-sequence (EOS) or initiate a sequencer reset, which brings the sequencer immediately back to the idle start state (CONV00 for SEQ1 and cascaded cases; CONV08 for SEQ2).
 - When SEQ1/2 are used in cascaded mode, triggers going to SEQ2 are ignored, while SEQ1 triggers are active. Cascaded mode can be viewed as SEQ1 with 16 states instead of eight.
-

1.3.4 Interrupt Operation During Sequenced Conversions

The sequencer can generate interrupts under two operating modes. These modes are determined by the Interrupt-Mode-Enable control bits in ADCTRL2.

A variation of [Example 1-4](#) can be used to show how interrupt mode 1 and mode 2 are useful under different operating conditions.

Case 1: Number of samples in the first and second sequences are not equal

- Mode 1 Interrupt operation (i.e., Interrupt request occurs at every EOS)
 1. Sequencer is initialized with MAX_CONVn = 1 for converting I_1 and I_2
 2. At ISR "a", MAX_CONVn is changed to 2 (by software) for converting V_1 , V_2 , and V_3
 3. At ISR "b", the following events take place :
 - a. MAX_CONVn is changed to 1 again for converting I_1 and I_2 .
 - b. Values I_1 , I_2 , V_1 , V_2 , and V_3 are read from ADC result registers.
 - c. The sequencer is reset.
 4. Steps 2 and 3 are repeated. Note that the interrupt flag is set every time SEQ_CNTR reaches zero and both interrupts are recognized.

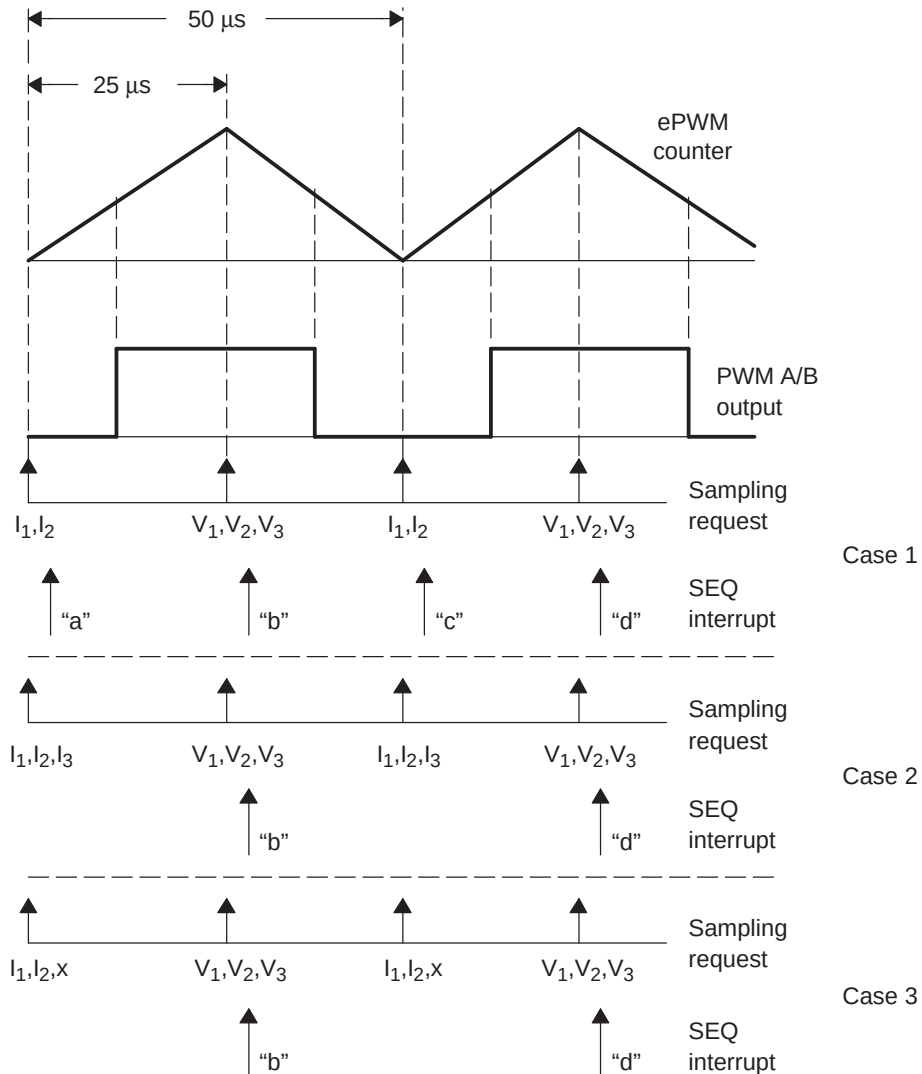
Case 2: Number of samples in the first and second sequences are equal

- Mode 2 Interrupt operation (i.e., Interrupt request occurs at every other EOS)
 1. Sequencer is initialized with MAX_CONVn = 2 for converting I_1 , I_2 , and I_3 (or V_1 , V_2 , and V_3).
 2. At ISR "b" and "d", the following events take place :
 - a. Values I_1 , I_2 , I_3 , V_1 , V_2 , and V_3 are read from ADC result registers.
 - b. The sequencer is reset.
 3. Step 2 is repeated.

Case 3: Number of samples in the first and second sequences are equal (with dummy read)

- Mode 2 Interrupt operation (i.e., Interrupt request occurs at every other EOS)
 1. Sequencer is initialized with MAX_CONVn = 2 for I_1 , I_2 , and x(dummy sample).
 2. At ISR "b" and "d", the following events take place :
 - a. Values I_1 , I_2 , x, V_1 , V_2 , and V_3 are read from ADC result registers.
 - b. The sequencer is reset.
 3. Step 2 is repeated. Note that the third I-sample (x) is a dummy sample, and is not really required. However, to minimize ISR overhead and CPU intervention, advantage is taken of the "every other" Interrupt request feature of Mode 2.

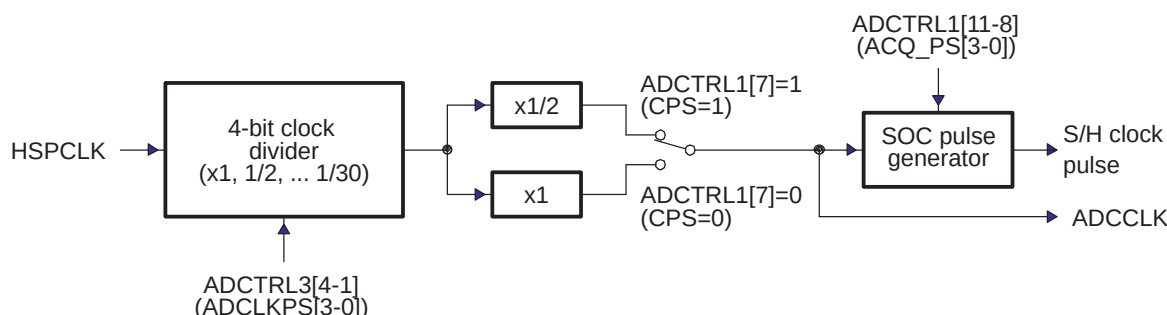
Figure 1-8. Interrupt Operation During Sequenced Conversions



1.4 ADC Clock Prescaler

The peripheral clock HSPCLK is divided down by the ADCCLKPS[3:0] bits of the ADCTRL3 register. An extra divide-by-two is provided via the CPS bit of the ADCTRL1 register. In addition, the ADC can be tailored to accommodate variations in source impedances by widening the sampling/acquisition period. This is controlled by the ACQ_PS[3:0] bits in the ADCTRL1 register. These bits do not affect the conversion portion of the S/H and conversion process, but do extend the length of time in which the sampling portion takes by extending the start of the conversion pulse. See [Figure 1-9](#).

Figure 1-9. ADC Core Clock and Sample-and-Hold (S/H) Clock



- A See register bit definition for clock divider ratio and S/H pulse control. S/H pulse width determines the size of acquisition window (the time period for which sampling switch is closed).

1.4.1 ADC-module Clock and Sample Rate

The ADC module has several prescaler stages to generate any desired ADC operating clock speed. [Figure 1-10](#) defines the clock selection stages that feed the ADC module. [Table 1-7](#) gives two example settings and shows both the effective sustained sequential sampling rate and the sample and hold window time for those settings.

Figure 1-10. Clock Chain to the ADC

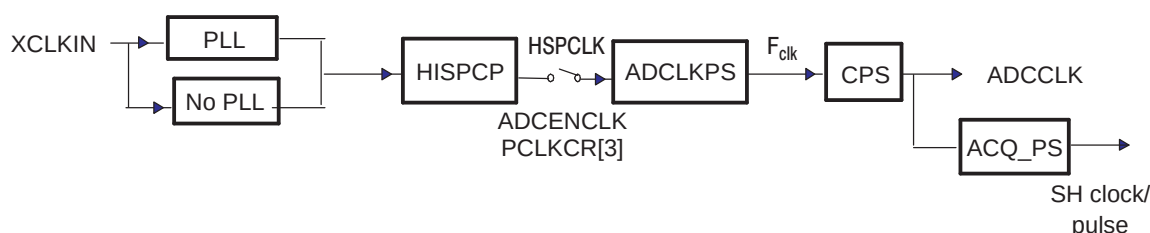


Table 1-7. Clock Chain to the ADC

XCLKIN	SYSCLOCKOUT	HSPCLK	ADCTRL3[4-1] ADCLKPS[3-0]	ADCTRL1[7] CPS	ADCCLK	ADCTRL1[11-8] ACQ_PS[3-0]	SH Width
30 MHz	150 MHz	HSPCLK = 3 150 MHz/ 2 X 3 = 25 MHz	ADCLKPS = 0 25 MHz	CPS=0 25 MHz	25 MHz	ACQ_PS = 0 12.5 MHz 12.5 MSPS sustained conversion rate	1 ADC Clock 40 ns
20 MHz	100 MHz	HSPCLK = 2 100 MHz/ 2 X 2 = 25 MHz	ADCLKPS = 2 25/2 X 2 = 6.25 MHz	CPS = 1 6.25 MHz/ 2 X 1 = 3.125 MHz	3.125 MHz	ACQ_PS = 15 183.824 kHz 183.824 kSPS sustained conversion rate	16 ADC Clocks 5.12 µs

1.5 Low-power Modes

The ADC supports three separate power sources each controlled by independent bits in the ADCTRL3 register. These three bits combine to make up three power levels: ADC power up, ADC power down, and ADC off.

Table 1-8. Power Options

Power Level	ADCBGRFDN1	ADCBGRFDN0	ADCPWDN
ADC power-up	1	1	1
ADC power-down	1	1	0
ADC off	0	0	0
Reserved	1	0	X
Reserved	0	1	X

1.6 Power-up Sequence

The ADC resets to the ADC off state. When powering up the ADC, use the following sequence:

1. If external reference is desired, enable this mode using bits 15-14 in the ADCREFSEL Register. This mode must be enabled before band gap is powered.
2. Power up the reference, bandgap, and analog circuits together by setting bits 7-5 (ADCBGRFDN[1:0], ADCPWDN) in the ADCTRL3 register.
3. Before performing the first conversion, a delay of 5 ms is required.

When powering down the ADC, all three bits can be cleared simultaneously. The ADC power level must be controlled via software and they are independent of the state of the device power modes.

Sometimes it is desirable to power down the ADC while leaving the band-gap and reference powered by clearing the ADCPWDN bit only. When the ADC is re-powered, a delay of 20 μ s is required after this bit is set before performing any conversions.

Note: The 2833x ADC requires a 5-ms delay after all of the circuits are powered up. This differs from the 281x ADC.

1.7 Sequencer Override Feature

In normal operation, sequencers SEQ1, SEQ2 or cascaded SEQ1 help to convert selected ADC channels and store them in the respective ADCRESULTn registers, sequentially. The sequence naturally wraps around at the end of the MAX_CONVn setting. With the sequencer override feature, the natural wraparound of the sequencers can be controlled in software. The sequencer override feature is controlled by bit 5 of the ADC Control Register 1 (ADCCTRL1).

For example, assume the SEQ_OVRD bit is 0 and the ADC is in cascaded-sequencer, continuous-conversion mode with MAX_CONV1 set to 7. Normally, the sequencer would increment sequentially and update up to ADCRESULT7 register with ADC conversions and wraps around to 0. At the end of the ADCRESULT7 register update, the relevant interrupt flag would be set.

With the SEQ_OVRD bit set to 1, the sequencer updates seven result registers and does *not* wrap around to 0. Instead, the sequencer will increment sequentially and update the ADCRESULT8 register onwards until the ADCRESULT15 register is reached. After updating ADCRESULT15 register, the natural wrap around to 0 will occur. This feature treats the result registers (0-15) like a FIFO for sequential data capture from the ADC. This feature is very helpful to capture ADC data when ADC conversions are done at the maximum data rate.

Recommendations and caution on sequencer override feature:

- After reset, SEQ_OVRD bit will be 0; therefore the sequencer override feature remains disabled.
- When SEQ_OVRD bit is set for all nonzero values of MAX_CONVn, the related interrupt flag bit will be set for every MAX_CONVn count of result register update.
- For example, if ADCMAXCONV is set to 3, then the interrupt flag for the selected sequencer will be set every four result register updates. The wraparound always occurs at the end of the sequencer (i.e., after ADCRESULT15 register update in cascaded sequencer mode).
- This will be functional in conversions using SEQ1, SEQ2, and cascaded sequencers using SEQ1.
- It is recommended that this feature not be enabled/controlled dynamically within the program. Always enable this feature during the ADC module initialization.
- In continuous-conversion mode with sequencer changes, the ADC channel address uses the preset values in CONVxx registers. If continuous conversions of the same channel are needed then all the CONVxx registers should have the same channel address.
- In continuous-conversion mode, if a sequencer reset is needed: set CONT_RUN bit to 0, wait 2 cycles in the ADC Clock domain, then reset the sequencer. CONT_RUN can then be set back to 1.
- For example, to get 16 contiguous samples for the ADCINA0 channel using the sequencer override feature, all 16 CONVxx registers should be set to 0x0000.

1.8 ADC Calibration

The ADC_cal() routine is programmed into TI reserved OTP memory by the factory. The boot ROM automatically calls the ADC_cal() routine to initialize the ADCREFSEL and ADCOFFTRIM registers with device specific calibration data. During normal operation, this process occurs automatically and no action is required by the user.

If the boot ROM is bypassed by Code Composer Studio during the development process, then ADCREFSEL and ADCOFFTRIM must be initialized by the application. For working examples, see the ADC initialization in the *C2833x C/C++ Header Files and Peripheral Examples* ([SPRC530](#)).

The next two sections describe different methods for calling the ADC_Cal() routine.

Note: FAILURE TO INITIALIZE THESE REGISTERS WILL CAUSE THE ADC TO FUNCTION OUT OF SPECIFICATION.

Because TI reserved OTP memory is secure, the ADC_Cal() routine must be called from secure memory or called from non-secure memory after the Code Security Module is unlocked. If the system is reset or the ADC module is reset using Bit 14 (RESET) from the ADC Control Register 1, the routine must be repeated.

1.8.1 ADC_Cal Assembly Routine Method

The following three steps describe how to call the ADC_cal routine from an application:

- Step 1. Add the ADC_cal assembly function to your project. The source is included with the C2833x C/C++ Header Files and Peripheral Examples ([SPRC530](#)). The following code shows the contents of the ADC_cal function. The values 0xAAAA and 0xB BBB are place holders. The actual values programmed by TI are device specific.

```

;-----
; This is the ADC cal routine. This routine is programmed
; into reserved memory by the factory. 0xAAAA and 0xB BBB
; are place holders. The actual values programmed by TI
; are device specific.
; The ADC clock must be enabled before calling
; this function.
;-----
.def _ADC_cal
.asg "0x711C", ADCREFSEL_LOC
.sect ".adc_cal"
_ADC_cal
MOVW DP, #ADCREFSEL_LOC >> 6
MOV @28, #0xAAAA
MOV @29, #0xB BBB
LRETR

```

- Step 2. Add the .adc_cal section to your linker command file using the following:

```

MEMORY
{
PAGE 0 :
...
ADC_CAL : origin = 0x380080, length = 0x000009
...
}
SECTIONS
{
...
.adc_cal : load = ADC_CAL, PAGE = 0, TYPE = NOLOAD
...
}

```

- Step 3. Call the ADC_cal function before using the ADC. The ADC clocks must be enabled before making this call.

```

extern void ADC_cal(void);
...
EALLOW;
SysCtrlRegs.PCLKCR0.bit.ADCENCLK = 1;
ADC_cal();
SysCtrlRegs.PCLKCR0.bit.ADCENCLK = 0;
EDIS;

```

1.8.2 Pointer to-Function Method

Because the ADC_CAL() routine is already programmed to TI reserved OTP memory by the factory, it can be called via a pointer to the function by using the following steps:

- Step 1. Define ADC_Cal as a pointer to the programmed function in the OTP memory.

```
#define ADC_Cal (void (*)(void)) 0x380080
```

- Step 2. Call the ADC_Cal() function.

```

...
EALLOW;
SysCtrlRegs.PCLKCR0.bit.ADCENCLK=1;
(*ADC_Cal) ();
SysCtrlRegs.PCLKCR0.bit.ADCENCLK=0;
EDIS;

```

1.9 Internal/External Reference Voltage Selection

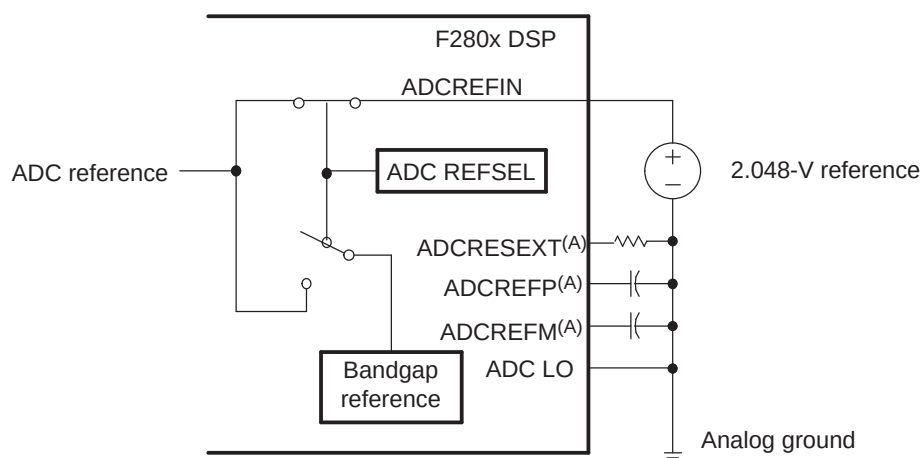
By default, an internally generated bandgap voltage reference is selected to supply the ADC logic.

Based on customer application requirements, the ADC logic may be supplied by an external voltage reference. The ADC will accept 2.048 V, 1.5 V, or 1.024 V on the ADCREFIN pin. The value of the ADCREFSEL register determines the reference source selected.

If the internal reference option is chosen, the ADCREFIN pin can be left connected to the selected source, left floating, or grounded. Regardless of which option is chosen, the external circuit for the ADCRESEXT, ADCREFP, and ADCREFM pins is the same.

The external reference voltage of 2.048 V was chosen to match industry standard reference components. These components are available in various temperature ratings. A recommended Texas Instruments part is REF3020AIDBZ.

Figure 1-11. External Bias for 2.048-V External Reference



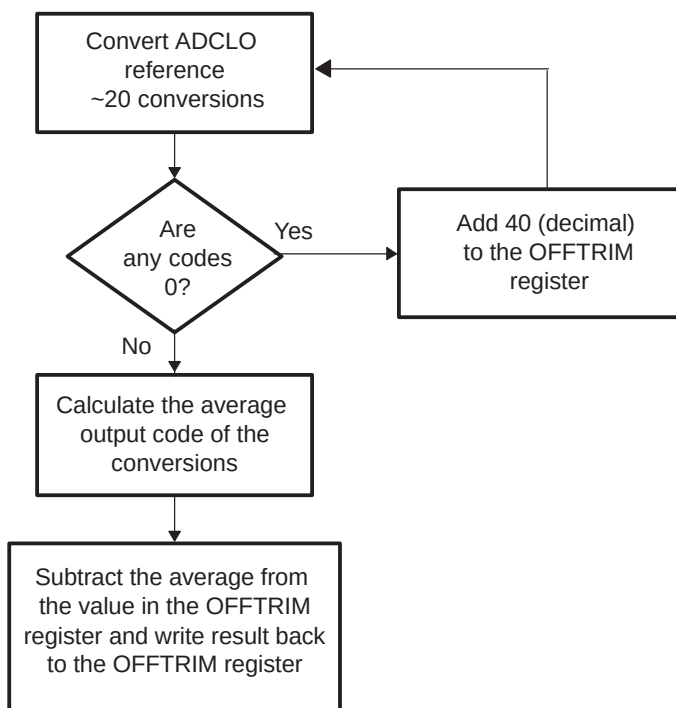
- A For component values, see the *TMS320F28332, TMS320F28334, TMS320F28335 Digital Signal Controllers (DSCs) Data Manual* (literature number SPRS439).

1.10 Offset Error Correction

The 2833x ADC supports offset correction via a 9-bit field in the ADC Offset Trim Register(ADCOFFTRIM). The value contained in this register will be added/subtracted before the results are available in the ADC result registers. This operation is contained in the ADC module, so timing for results will not be affected. Furthermore, since the operation is handled inside the ADC, the full dynamic range of the ADC will be maintained for any trim value.

The ADCOFFTRIM register is pre-loaded by the ADC_cal routine in the boot ROM. To further reduce the offset error in the target application, connect the signal ADCLO to one of the ADC channels and convert that channel, modifying the value in the ADCOFFTRIM, until a centered zero code is observed. See [Figure 1-12](#) for a flow diagram.

Figure 1-12. Flow Chart of Offset Error Correction Process



Example 1-5. Negative Offset

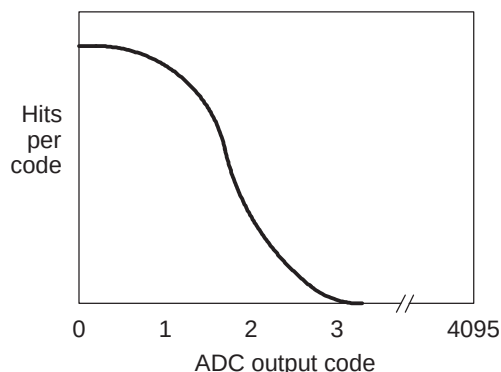
At startup, most of the reference conversions yield a zero result. After writing the value 0x28 (40 decimal) into the OFFTRIM register, all of the reference conversions give a positive result and average out to 0x19 (25 decimal). The final value written to the OFFTRIM register should be 0x0F (15 decimal).

Example 1-6. Positive Offset

At startup, all of the reference conversions yield a positive result with an average of 0x14 (20 decimal). The final value written to the OFFTRIM register should be 0x1EC (-20 decimal).

After the offset error correction process is completed, a half bell curved distribution similar to [Figure 1-13](#) should be seen when multiple ADCLO samples are converted. The other half of the bell curve is hidden due to the fact that the converter bottoms out at a code of zero.

Figure 1-13. Ideal Code Distribution of Sampled 0-V Reference



1.11 ADC to DMA Interface

The ADC result registers located in peripheral frame 0 (0x0B00 – 0x0B0F) are accessible by the DMA unit on the F2833x. These registers can also be accessed by the CPU at the same time as the DMA without bus contention. The result registers in peripheral frame 2 (0x7108 – 0x710F) are not accessible by the DMA.

There is a sync signal provided automatically by the ADC to the DMA for a sequencer 1 conversion when both SEQ_OVRD and CONT_RUN bits are set. The sync pulse will be generated by the ADC after the first MAXCONV limit is reached for each pass through the sequencer. When the sequencer 1 is in this configuration it is possible that the DMA could become misaligned to the currently populated result registers, depending on the loading of the other DMA channels. If a misalignment occurs, the DMA can use the sync signal to detect and flag a sync error event.

For more information on how the sync signal is used locally in the DMA, please see the *TMS320F2833x Direct Memory Access (DMA) Reference Guide* (literature number [SPRUFB8](#)).

ADC Registers

This chapter contains the ADC registers and bit definitions, with the registers grouped by function.

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2.3 Autosequence Status Register (ADCSEQSR)	40
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2.1 ADC Control Registers

Figure 2-1. ADC Control Register 1 (ADCTRL1) (Address Offset 00h)

15	14	13	12	11	8
Reserved	RESET	SUSMOD		ACQ_PS	
R-0	R/W-0	R/W-0		R/W-0	
7	6	5	4	3	0
CPS	CONT_RUN	SEQ_OVRD	SEQ_CASC		Reserved
R/W-0	R/W-0	R/W-0	R/W-0		R-0

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-1. ADC Control Register 1 (ADCTRL1) Field Descriptions

Bit(s)	Name	Value	Description
15	Reserved		Reads return a zero. Writes have no effect.
14	RESET	0 No effect 1 Resets entire ADC module (bit is then set back to 0 by ADC logic)	ADC module software reset. This bit causes a master reset on the entire ADC module. All register bits and sequencer state machines are reset to the initial state as occurs when the device reset pin is pulled low (or after a power-on reset). This is a one-time-effect bit, meaning this bit is self-cleared immediately after it is set to 1. Read of this bit always returns a 0. Also, the reset of ADC has a latency of two clock cycles (that is, other ADC control register bits should not be modified until two ADC clock cycles after the instruction that resets the ADC). Note: The ADC module is reset during a system reset. If an ADC module reset is desired at any other time, you can do so by writing a 1 to this bit. After two ADC clock domain cycles, you can then write the appropriate values to the ADCTRL1 register bits. The example below assumes 150-MHz DSP Clock and 25-MHz ADCCLK. Assembly code: MOV ADCTRL1, #01xxxxxxxxxxxxxb; Resets the ADC (RESET = 1) RPT #10 NOP; Provides the required delay between writes to ADCTRL1 MOV ADCTRL1, #00xxxxxxxxxxxxxb; Configures ADCTRL1 to user-desired value. Note that the second MOV is not required if the default configuration is sufficient.
13-12	SUSMOD[1:0]	00 Mode 0. Emulation suspend is ignored. 01 Mode 1. Sequencer and other wrapper logic stops after current sequence is complete, final result is latched, and state machine is updated. 10 Mode 2. Sequencer and other wrapper logic stops after current conversion is complete, result is latched, and state machine is updated. 11 Mode 3. Sequencer and other wrapper logic stops immediately on emulation suspend.	Emulation-suspend mode. These bits determine what occurs when an emulation-suspend occurs (due to the debugger hitting a breakpoint, for example).
11-8	ACQ_PS[3:0]		Acquisition window size. This bit field controls the width of SOC pulse, which, in turn, determines for what time duration the sampling switch is closed. The width of SOC pulse is ADCTRL1[11:8] + 1 times the ADCLK period.
7	CPS	0 ADCCLK = F _{clk} /1 1 ADCCLK = F _{clk} /2	Core clock prescaler. The prescaler is applied to divided device peripheral clock, HSPCLK. Note: F_{clk} = Prescaled HSPCLK (ADCCLKPS[3:0])
6	CONT_RUN		Continuous run. This bit determines whether the sequencer operates in continuous conversion mode or start-stop mode. This bit can be written while a current conversion sequence is active. This bit will take effect at the end of the current conversion sequence; i.e., software can set/clear this bit until EOS has occurred, for valid action to be taken. In the continuous conversion mode, there is no need to reset the sequencer; however, the sequencer must be reset in the start-stop mode to put the converter in state CONV00.

Table 2-1. ADC Control Register 1 (ADCTRL1) Field Descriptions (continued)

Bit(s)	Name	Value	Description
		0	Start-stop mode. Sequencer stops after reaching EOS. On the next SOC, the sequencer starts from the state where it ended unless a sequencer reset is performed.
		1	Continuous conversion mode. After reaching EOS, the behavior of the sequencer depends on the state of the SEQ_OVRD bit. If this bit is cleared, the sequencer starts over again from its reset state (CONV00 for SEQ1 and cascaded, CONV08 for SEQ2). If SEQ_OVRD is set, the sequencer starts again from its current position, without resetting.
5	SEQ_OVRD	0	Sequencer override. Provides additional sequencer flexibility in continuous run mode by overriding the wrapping around at the end of conversions set by MAX_CONVn.
		1	Disabled - Allows the sequencer to wrap around at the end of conversions set by MAX_CONVn.
		1	Enabled - Overrides the sequencer from wrapping around at the end of conversions set by MAX_CONVn. Wraparound occurs only at the end of the sequencer.
4	SEQ_CASC	0	Cascaded sequencer operation. This bit determines whether SEQ1 and SEQ2 operate as two 8-state sequencers or as a single 16-state sequencer (SEQ).
		0	Dual-sequencer mode. SEQ1 and SEQ2 operate as two 8-state sequencers.
		1	Cascaded mode. SEQ1 and SEQ2 operate as a single 16-state sequencer (SEQ).
3-0	Reserved		Reads return zero. Writes have no effect.

Figure 2-2. ADC Control Register 2 (ADCTRL2) (Address Offset 01h)

15	14	13	12	11	10	9	8
ePWM_SOCB_SEQ	RST_SEQ1	SOC_SEQ1	Reserved	INT_ENA_SEQ1	INT_MOD_SEQ1	Reserved	ePWM_SOCB_SEQ1
R/W-0	R/W-0	R/W-0	R-0	R/W-0	R/W-0	R-0	R/W-0
7	6	5	4	3	2	1	0
EXT_SOC_SEQ1	RST_SEQ2	SOC_SEQ2	Reserved	INT_ENA_SEQ2	INT_MOD_SEQ2	Reserved	ePWM_SOCB_SEQ2
R/W-0	R/W-0	R/W-0	R-0	R/W-0	R/W-0	R-0	R/W-0

LEGEND: R/W = Read/Write; R = Read only; W = Write access, S = Set only, C = Clear -n = value after reset

Table 2-2. ADC Control Register 2 (ADCTRL2) Field Descriptions

Bit(s)	Name	Value	Description
15	ePWM_SOCB_SEQ	0	ePWM SOCB enable for cascaded sequencer (<i>Note: This bit is active only in cascaded mode.</i>)
		1	No action
		1	Setting this bit allows the cascaded sequencer to be started by an ePWM SOCB signal. The ePWM modules can be programmed to start a conversion on various events. See the <i>TMS320x28xx, 28xxx Enhanced Pulse Width Modulation Module Reference Guide</i> (literature number SPRU791) for more information on the ePWM modules.
14	RST_SEQ1	0	Reset sequencer1 Writing a 1 to this bit resets SEQ1 or the cascaded sequencer immediately to an initial "pretriggered" state, i.e., waiting for a trigger at CONV00. A currently active conversion sequence will be aborted.
		0	No action
		1	Immediately reset sequencer to state CONV00

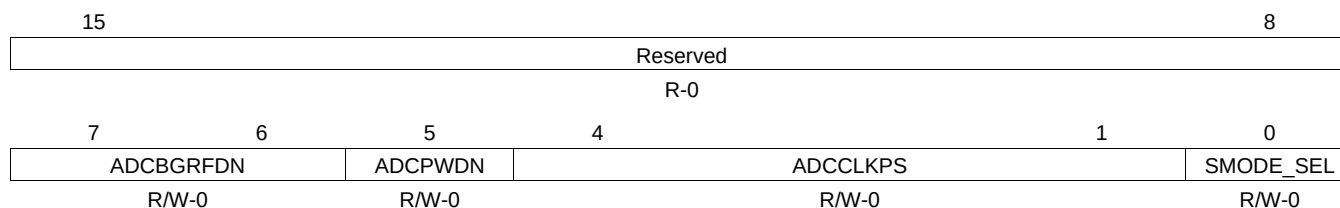
Table 2-2. ADC Control Register 2 (ADCTRL2) Field Descriptions (continued)

Bit(s)	Name	Value	Description
13	SOC_SEQ1	0 1	Start-of-conversion (SOC) trigger for Sequencer 1 (SEQ1) or the cascaded sequencer. This bit can be set by the following triggers: • S/W - Software writing a 1 to this bit • ePWM SOCA • ePWM SOCB (only in cascaded mode) • EXT - External pin (i.e., GPIO Port A pin (GPIO31-0) configured as XINT2 in the GPIOxINT2SEL register. See the <i>TMS320x2833x System Control and Interrupts Reference Guide</i> (literature number SPRUFB0) for details on how to configure a GPIO pin as XINT2. When a trigger occurs, there are three possibilities: Case 1: SEQ1 idle and SOC bit clear SEQ1 starts immediately (under arbiter control). This bit is set and cleared, allowing for any "pending" trigger requests. Case 2: SEQ1 busy and SOC bit clear Bit is set signifying a trigger request is pending. When SEQ1 finally starts after completing current conversion, this bit is cleared. Case 3: SEQ1 busy and SOC bit set Any trigger occurring in this case is ignored (lost). Clears a pending SOC trigger. Note: If the sequencer has already started, this bit is automatically cleared, and hence, writing a zero has no effect; i.e., an already started sequencer cannot be stopped by clearing this bit. Software trigger - Start SEQ1 from currently stopped position (i.e., Idle mode) Note: The RST_SEQ1 (ADCTRL2.14) and the SOC_SEQ1 (ADCTRL2.13) bits should not be set in the same instruction. This resets the sequencer, but does not start the sequence. The correct sequence of operation is to set the RST_SEQ1 bit first, and the SOC_SEQ1 bit in the following instruction. This makes certain that the sequencer is reset and a new sequence started. This sequence applies to the RST_SEQ2 (ADCTRL2.6) and SOC_SEQ2 (ADCTRL2.5) bits also.
12	Reserved		Reads return a zero. Writes have no effect.
11	INT_ENA_SEQ1	0 1	SEQ1 interrupt enable. This bit enables the interrupt request to CPU by INT_SEQ1. Interrupt request by INT_SEQ1 is disabled. Interrupt request by INT_SEQ1 is enabled.
10	INT_MOD_SEQ1	0 1	SEQ1 interrupt mode. This bit selects SEQ1 interrupt mode. It affects the setting of INT_SEQ1 at the end of the SEQ1 conversion sequence. INT_SEQ1 is set at the end of every SEQ1 sequence. INT_SEQ1 is set at the end of every other SEQ1 sequence.
9	Reserved		Reads return a zero. Writes have no effect.
8	ePWM_SOCA_SEQ1	0 1	ePWM SOCA enable bit for SEQ1 SEQ1 cannot be started by ePWMx SOCA trigger. Allows SEQ1/SEQ to be started by ePWMx SOCA trigger. The ePWMs can be programmed to start a conversion on various events.
7	EXT_SOC_SEQ1	0 1	External signal start-of-conversion bit for SEQ1 No action Setting this bit enables an ADC autoconversion sequence to be started by a signal from a GPIO Port A pin (GPIO31-0) configured as XINT2 in the GPIOXINT2SEL register. See the <i>TMS320x2833x System Control and Interrupts Reference Guide</i> (literature number SPRUFB0)
6	RST_SEQ2	0 1	Reset SEQ2 No action Immediately resets SEQ2 to an initial "pretriggered" state, i.e., waiting for a trigger at CONV08. A currently active conversion sequence will be aborted.

Table 2-2. ADC Control Register 2 (ADCTRL2) Field Descriptions (continued)

Bit(s)	Name	Value	Description
5	SOC_SEQ2	0 1	Start of conversion trigger for sequencer 2 (SEQ2). <i>(Only applicable in dual-sequencer mode; ignored in cascaded mode.)</i> This bit can be set by the following triggers: • S/W - Software writing of 1 to this bit • ePWM SOCB When a trigger occurs, there are three possibilities: Case 1: SEQ2 idle and SOC bit clear SEQ2 starts immediately (under arbiter control) and the bit is cleared, allowing for any pending trigger requests. Case 2: SEQ2 busy and SOC bit clear Bit is set signifying a trigger request is pending. When SEQ2 finally starts after completing current conversion, this bit will be cleared. Case 3: SEQ2 busy and SOC bit set Any trigger occurring in this case will be ignored (lost). Note: If the sequencer has already started, this bit is automatically cleared, and writing a zero has no effect; i.e., an already started sequencer cannot be stopped by clearing this bit. Clears a Pending SOC trigger Starts SEQ2 from currently stopped position (i.e., Idle mode)
4	Reserved		Reads return a zero. Writes have no effect.
3	INT_ENA_SEQ2	0 1	SEQ2 interrupt enable. This bit enables or disables an interrupt request to the CPU by INT_SEQ2. Interrupt request by INT_SEQ2 is disabled. Interrupt request by INT_SEQ2 is enabled.
2	INT_MOD_SEQ2	0 1	SEQ2 interrupt mode. This bit selects SEQ2 interrupt mode. It affects the setting of INT_SEQ2 at the end of the SEQ2 conversion sequence. INT_SEQ2 is set at the end of every SEQ2 sequence. INT_SEQ2 is set at the end of every other SEQ2 sequence.
1	Reserved		Reads return a zero. Writes have no effect.
0	ePWM_SOCB_SEQ2	0 1	ePWM SOCB enable bit for SEQ2. SEQ2 cannot be started by ePWMx SOCB trigger. Allows SEQ2 to be started by ePWMx SOCB trigger. The ePWMs can be programmed to start a conversion on various events.

Figure 2-3. ADC Control Register 3 (ADCTRL3) (Address Offset 18h)



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

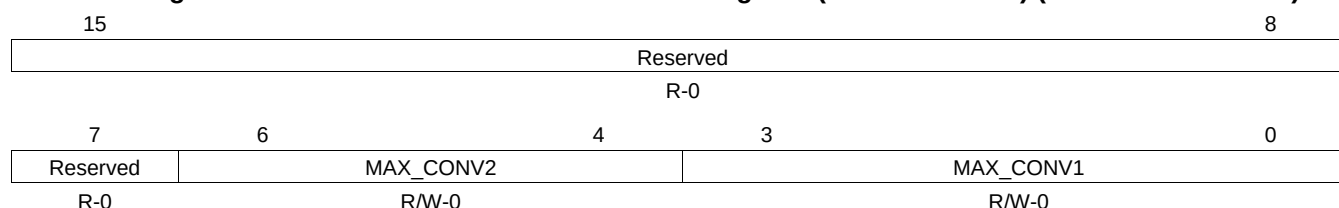
Table 2-3. ADC Control Register 3 (ADCTRL3) Field Descriptions

Bit(s)	Name	Value	Description
15-8	Reserved		Reads return a zero. Writes have no effect.
7-6	ADCBGRFDN[1:0]	00 11	ADC bandgap and reference power down. These bits control the power up and power down of the bandgap and reference circuitry inside the analog core. See Section 1.6 for power-up sequence requirements. The bandgap and reference circuitry is powered down. The bandgap and reference circuitry is powered up.
5	ADCPWDN		ADC power down. This bit controls the power up and power down of all the analog circuitry inside the analog core except the bandgap and reference circuitry. See Section 1.6 for power-up sequence requirements.

Table 2-3. ADC Control Register 3 (ADCTRL3) Field Descriptions (continued)

Bit(s)	Name	Value	Description
		0	All analog circuitry inside the core except the bandgap and reference circuitry is powered down.
		1	The analog circuitry inside the core is powered up.
4-1	ADCCLKPS [3:0]	0000 0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100 1101 1110 1111	Core clock divider. 28x peripheral clock, HSPCLK, is divided by 2*ADCCLKPS[3-0], except when ADCCLKPS[3-0] is 0000, in which case HSPCLK is directly passed on. The divided clock is further divided by ADCTRL1[7]+1 to generate the core clock, ADCLK. ADCCLKPS [3:0] Core Clock Divider ADCLK 0 HSPCLK/(ADCTRL1[7] + 1) 1 HSPCLK/[2*(ADCTRL1[7] + 1)] 2 HSPCLK/[4*(ADCTRL1[7] + 1)] 3 HSPCLK/[6*(ADCTRL1[7] + 1)] 4 HSPCLK/[8*(ADCTRL1[7] + 1)] 5 HSPCLK/[10*(ADCTRL1[7] + 1)] 6 HSPCLK/[12*(ADCTRL1[7] + 1)] 7 HSPCLK/[14*(ADCTRL1[7] + 1)] 8 HSPCLK/[16*(ADCTRL1[7] + 1)] 9 HSPCLK/[18*(ADCTRL1[7] + 1)] 10 HSPCLK/[20*(ADCTRL1[7] + 1)] 11 HSPCLK/[22*(ADCTRL1[7] + 1)] 12 HSPCLK/[24*(ADCTRL1[7] + 1)] 13 HSPCLK/[26*(ADCTRL1[7] + 1)] 14 HSPCLK/[28*(ADCTRL1[7] + 1)] 15 HSPCLK/[30*(ADCTRL1[7] + 1)]
0	SMODE_SEL	0 1	Sampling mode select. This bit selects either sequential or simultaneous sampling mode. 0 Sequential sampling mode is selected. 1 Simultaneous sampling mode is selected.

2.2 Maximum Conversion Channels Register (ADCMAXCONV)

Figure 2-4. Maximum Conversion Channels Register (ADCMAXCONV) (Offset Address 02h)


LEGEND: R/W = Read/Write; R = Read only; x = undefined, -n = value after reset

Table 2-4. Maximum Conversion Channels Register (ADCMAXCONV) Field Descriptions

Bit(s)	Name	Description
15-7	Reserved	Reads return a zero. Writes have no effect.
6-0	MAX_CONVn	MAX_CONVn bit field defines the maximum number of conversions executed in an autoconversion session. The bit fields and their operation vary according to the sequencer modes (dual/cascaded). For SEQ1 operation, bits MAX_CONV1[2:0] are used. For SEQ2 operation, bits MAX_CONV2[2:0] are used. For SEQ operation, bits MAX_CONV1[3:0] are used. An autoconversion session always starts with the initial state and continues sequentially until the end state if allowed. The result buffer is filled in a sequential order. Any number of conversions between 1 and (MAX_CONVn +1) can be programmed for a session.

Example 2-1. ADCMAXCONV Register Bit Programming

If only five conversions are required, then MAX_CONVn is set to four.

Case 1: Dual mode SEQ1 and cascaded mode Sequencer goes from CONV00 to CONV04, and the five conversion results are stored in the registers Result 00 to Result 04 of the Conversion Result Buffer.

Case 2: Dual mode SEQ2 Sequencer goes from CONV08 to CONV12, and the five conversion results are stored in the registers Result 08 to Result 12 of the Conversion Result Buffer.

MAX_CONV1 Value >7 for Dual-Sequencer Mode

If a value for MAX_CONV1, which is greater than 7, is chosen for the dual-sequencer mode (i.e., two separate 8-state sequencers), then SEQ_CNTR will continue counting past seven, causing the sequencer to wrap around to CONV00 and continue counting.

Table 2-5. Bit Selections for MAX_CONV1 for Various Number of Conversions

ADCMAXCONV[3-0]	Number of Conversions
0000	1
0001	2
0010	3
0011	4
0100	5
0101	6
0110	7
0111	8
1000	9
1001	10
1010	11
1011	12
1100	13
1101	14
1110	15
1111	16

2.3 Autosequence Status Register (ADCASEQSR)

Figure 2-5. Autosequence Status Register (ADCASEQSR) (Address Offset 07h)

15	12	11	8
Reserved		SEQ_CNTR	
R-0		R-0	
7	6	4	3
Reserved	SEQ2_STATE		SEQ1_STATE
R-0	R-0		R-0

LEGEND: R/W = Read/Write; R =Read only; x = undefined, -n = value after reset

Table 2-6. Autosequence Status Register (ADCASEQSR) Field Descriptions

Bit(s)	Name	Description
15-12	Reserved	Reads return a zero. Writes have no effect.
11-8	SEQ_CNTR[3:0]	Sequencing counter status bits. The SEQ_CNTRn 4-bit status field is used by SEQ1, SEQ2, and the cascaded sequencer. SEQ2 is irrelevant in cascaded mode. The Sequencer Counter bit field, SEQ_CNTR[3:0], is initialized to the value in MAX_CONV at the start of a conversion sequence. After each conversion (or a pair of conversions in simultaneous sampling mode) in an auto conversion sequence, the Sequencer Counter decreases by 1. The SEQ_CNTR bits can be read at any time during the countdown process to check status of the sequencer. This value, together with the SEQ1 and SEQ2 busy bits, uniquely identifies the progress or state of the active sequencer at any point in time. See Table 2-7 .
7	Reserved	Reads return a zero. Writes have no effect.
6-0	SEQ2_STATE[2:0] and SEQ1_STATE[3:0]	SEQ2_STATE and SEQ1_STATE bit fields are the pointers of SEQ2 and SEQ1, respectively.

Table 2-7. State of Active Sequencer

SEQ_CNTR (read only)	Number of conversions remaining
0000	1 or 0, depending on the busy bit
0001	2
0010	3
0011	4
0100	5
0101	6
0110	7
0111	8
1000	9
1001	10
1010	11
1011	12
1100	13
1101	14
1110	15
1111	16

2.4 ADC Status and Flag Register (ADCST)

Figure 2-6. ADC Status and Flag Register (ADCST) (Address Offset 19h)

15															8																																								
Reserved																																																							
R-0																																																							
7							6							5							4							3							2							1							0						
EOS_BUF2							EOS_BUF1							INT_SEQ2_CLR							INT_SEQ1_CLR							SEQ2_BSY							SEQ1_BSY							INT_SEQ2							INT_SEQ1						
R-0							R-0							R/W-0							R/W-0							R-0							R-0							R-0							R-0						

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

This register is a dedicated status and flag register. The bits in this register are either read-only status or flag bits, or read-return-zero condition clearing bits.

Table 2-8. ADC Status and Flag Register (ADCST) Field Descriptions

Bit(s)	Name	Value	Description
15-8	Reserved		Reads return a zero. Writes have no effect.
7	EOS_BUF2		End of sequence buffer bit for SEQ2. This bit is not used and remains as zero in interrupt mode 0, i.e. when ADCTRL2[2]=0. In interrupt mode 1, i.e. when ADCTRL2[2]=1, it toggles on every end of sequence of SEQ2. This bit is cleared on device reset and is not affected by sequencer reset or clearing of the corresponding interrupt flag.
6	EOS_BUF1		End of sequence buffer bit for SEQ1. This bit is not used and remains as zero in interrupt mode 0, i.e. when ADCTRL2[10]=0. In interrupt mode 1, i.e. when ADCTRL2[10]=1, it toggles on every end of sequence of SEQ1. This bit is cleared on device reset and is not affected by sequencer reset or clearing of the corresponding interrupt flag.
5	INT_SEQ2_CLR	0 1	Interrupt clear bit. Read of this bit always returns 0. The clear action is a one-shot event following a write of 1 to this bit. Writing a zero to this bit has no effect. Writing a 1 to this bit clears the SEQ2 interrupt flag bit, INT_SEQ2. This bit does not affect the EOS_BUF2 bit.
4	INT_SEQ1_CLR	0 1	Interrupt clear bit. Read of this bit always returns 0. The clear action is a one-shot event following a write of 1 to this bit. Writing a zero to this bit has no effect. Writing a 1 to this bit clears the SEQ1 interrupt flag bit, INT_SEQ1. This bit does not affect the EOS_BUF1 bit.
3	SEQ2_BSY	0 1	SEQ2 busy status bit. SEQ2 is in idle, waiting for trigger. SEQ2 is in progress. Write to this bit has no effect.
2	SEQ1_BSY	0 1	SEQ1 busy status bit. Write to this bit has no effect. SEQ1 is in idle, waiting for trigger. SEQ1 is in progress.
1	INT_SEQ2	0 1	SEQ2 interrupt flag bit. Write to this bit has no effect. In interrupt mode 0, i.e. when ADCTRL2[2]=0, this bit is set on every end of sequence of Seq 2. In interrupt mode 1, i.e., when ADCTRL2[2]=1, this bit is set on an end of sequence of Seq 2 if EOS_BUF2 is set. No SEQ2 interrupt event. SEQ2 interrupt event occurred.
0	INT_SEQ1	0 1	SEQ1 interrupt flag bit. Write to this bit has no effect. In interrupt mode 0, i.e. when ADCTRL2[10]=0, this bit is set on every end of sequence of Seq 1. In interrupt mode 1, i.e., when ADCTRL2[10]=1, this bit is set on an end of sequence of Seq 1 if EOS_BUF1 is set. No SEQ1 interrupt event. SEQ1 interrupt event occurred.

2.5 ADC Reference Select Register (ADCREFSSEL)

Figure 2-7. ADC Reference Select Register (ADCREFSSEL) (Address Offset 1Ch)

15	14	13	0
REF_SEL	Reserved		
R/W-0	R/W-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-9. ADC Reference Select Register (ADCREFSSEL) Field Descriptions

Bit(s)	Name	Value	Description
15-14	REF_SEL[1:0]	00 01 10 11	Reference select bits for ADC voltage generation circuit options are listed below: Internal reference selected (default) External reference, 2.048 V on ADCREFIN External reference, 1.500 V on ADCREFIN External reference, 1.024 V on ADCREFIN
13-0	Reserved		These bits are reserved for reference calibration data loaded from the Boot ROM. All writes to the ADCREFSEL register should leave the contents of this bit field as is after population by the Boot ROM.

2.6 ADC Offset Trim Register (ADCOFFTRIM)

Figure 2-8. ADC Offset Trim Register (ADCOFFTRIM) (Address Offset 1Dh)

15	9	8	0
Reserved		OFFSET_TRIM	
R-0		R/W-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 2-10. ADC Offset Trim Register (ADCOFFTRIM) Field Descriptions

Bit(s)	Name	Description
15-9	Reserved	Reads return a zero. Writes have no effect.
8-0	OFFSET_TRIM[8:0]	Offset trim value in LSBs, two's complement format; - 256/255 range

2.7 ADC Input Channel Select Sequencing Control Registers

Figure 2-9. ADC Input Channel Select Sequencing Control Registers (ADCCHSELSEQ1) (Address Offset 03h)

15	12	11	8	7	4	3	0
CONV03	CONV02	CONV01	CONV00				
R/W-0	R/W-0	R/W-0	R/W-0				

LEGEND: R/W = Read/Write; -n = value after reset

Figure 2-10. ADC Input Channel Select Sequencing Control Registers (ADCCHSELSEQ2) (Address Offset 04h)

15	12	11	8	7	4	3	0
CONV07	CONV06	CONV05	CONV04				
R/W-0	R/W-0	R/W-0	R/W-0				

LEGEND: R/W = Read/Write; -n = value after reset

Figure 2-11. ADC Input Channel Select Sequencing Control Registers (ADCCHSELSEQ3) (Address Offset 05h)

15	12	11	8	7	4	3	0
CONV11	CONV10	CONV09	CONV08				
R/W-0	R/W-0	R/W-0	R/W-0				

LEGEND: R/W = Read/Write; -n = value after reset

Figure 2-12. ADC Input Channel Select Sequencing Control Registers (ADCCHSELSEQ4) (Address Offset 06h)

15	12	11	8	7	4	3	0
CONV15	CONV14	CONV13	CONV12				
R/W-0	R/W-0	R/W-0	R/W-0				

LEGEND: R/W = Read/Write; -n = value after reset

Each of the 4-bit fields, CONVxx, selects one of the 16 MUXed analog input ADC channels for an autosequenced conversion.

Table 2-11. CONVnn Bit Values and the ADC Input Channels Selected

CONVnn Value	ADC Input Channel Selected
0000	ADCINA0
0001	ADCINA1
0010	ADCINA2
0011	ADCINA3
0100	ADCINA4
0101	ADCINA5
0110	ADCINA6
0111	ADCINA7
1000	ADCINB0
1001	ADCINB1
1010	ADCINB2
1011	ADCINB3
1100	ADCINB4

Table 2-11. CONVnn Bit Values and the ADC Input Channels Selected (continued)

CONVnn Value	ADC Input Channel Selected
1101	ADCINB5
1110	ADCINB6
1111	ADCINB7

2.8 ADC Conversion Result Buffer Registers (ADCRESULTn)

In the cascaded sequencer mode, registers ADCRESULT8 through ADCRESULT15 holds the results of the ninth through sixteenth conversions. The ADCRESULTn registers are left justified when read from Peripheral Frame 2 (0x7108-0x7117) with two wait states and right justified when read from Peripheral Frame 0 (0x0B00-0x0B0F) with zero wait states.

Figure 2-13. ADC Conversion Result Buffer Registers (ADCRESULTn) - (Addresses 0x7108-0x7117)

15	14	13	12	11	10	9	8
D11	D10	D9	D8	D7	D6	D5	D4
R-0	R-0	R-0	R-0	R-0	R-0	R-0	R-0
7	6	5	4	3	0		
D3	D2	D1	D0	Reserved			
R-0	R-0	R-0	R-0	R-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Figure 2-14. ADC Conversion Result Buffer Registers (ADCRESULTn) - (Addresses 0x0B00-0x0B0F)

15				12		11		10		9		8			
Reserved								D11		D10		D9		D8	
R-0								R-0		R-0		R-0		R-0	
7		6		5		4		3		2		1		0	
D7		D6		D5		D4		D3		D2		D1		D0	
R-0		R-0		R-0		R-0		R-0		R-0		R-0		R-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

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Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265
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